

DIGITAL GAUGE And SUPER ELEVATION MEASURING System For Railway Tracks

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Railway track laying, construction, and maintenance demand precise measurement of the gauge between two parallel tracks and superelevation, that is, the height difference between rails. For long welded rail

tracks, track temperature monitoring is also critical, as rail behaviour is highly sensitive to weather-induced thermal variations. The availability of low-cost sensors and micro-controllers such as Arduino now enables the development of compact digital systems that perform these measurements automatically with improved accuracy.

This system uses an Arduino Nano as the main controller to measure and display track level, its gauge, and temperature. An MPU6050 sensor is employed for tilt (level) measurement, a VL53L0X laser distance sensor measures the track gauge or gap, and a DS18B20 sensor monitors rail track

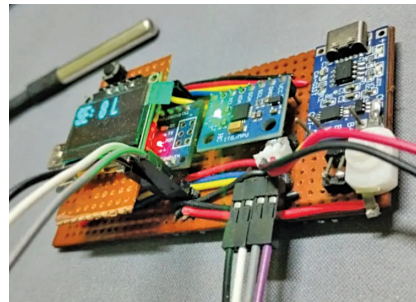


Fig. 1: Author's prototype

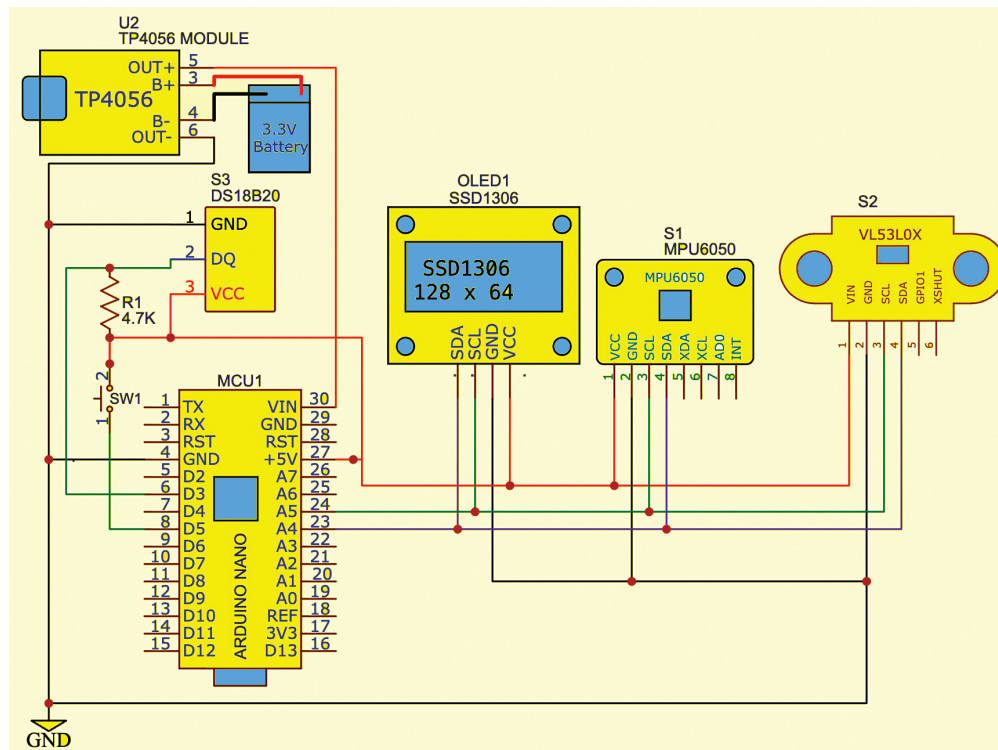


Fig. 2: Circuit diagram

BILL OF MATERIALS

Components	Quantity
Arduino Nano (MCU1)	1
SSD1306 OLED display (OLED1)	1
MPU6050 IMU sensor (S1)	1
VL53L0X TOF sensor (S2)	1
DS18B20 sensor (S3)	1
3.7V battery	1
TP4056 battery charger and power module (U2)	1
4.7k resistor (R1)	1
Push-to-on switch (SW1)	1

temperature. All measured parameters are clearly displayed on an OLED display, while a single push-button enables switching between different measurement modes.

Powered by a small battery, the system is fully portable and well suited for field use. It is intended for hobbyists, field engineers, and DIY enthusiasts who require a simple, reliable, and accurate electronic measuring instrument. Fig. 1 shows the author's prototype, and the required components are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the digital gauge and super elevation measuring system. The circuit is built around an Arduino Nano (MCU1), a TP4056 module (U2),